A Few Notes on Vector Calculus

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Last update: March 15, 2022

1 Introduction

TBD

2 Notation

Some of the notation we'll use is described in this section.

2.1 Vectors

Some authors use bold font to denote a vector, e.g. $\mathbf{u}$. Adding a "hat" denotes a unit vector. That is, for a vector $\mathbf{u}$, $\hat{\mathbf{u}}$ is defined to be

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

In words: $\hat{\mathbf{u}}$ is a vector of unit length in the $\mathbf{u}$ direction.

Similarly, $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{i}}$ are unit vectors in $\mathbb{R}^3$ in the x, y, and z directions respectively. Note that $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are the canonical basis vectors for $\mathbb{R}^3$ [2] and have column vector format

$$\hat{\mathbf{i}} = \begin{bmatrix}1\\0\\0\end{bmatrix},\ \hat{\mathbf{j}} = \begin{bmatrix}0\\1\\0\end{bmatrix},\ \hat{\mathbf{k}} = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

I have also seen $\mathbf{e}_i$ used to represent the ith basis vector in $\mathbb{R}^n$. So $\mathbf{e}_i$ is a vector with a one in the ith position and a zero in each of the other $n-1$ positions. In $\mathbb{R}^3$ this means that $\mathbf{e}_1 = \hat{\mathbf{i}}$, $\mathbf{e}_2 = \hat{\mathbf{j}}$, and $\mathbf{e}_3 = \hat{\mathbf{k}}$.

In general, the standard basis (which is sometimes called the computational basis) for the n-dimensional Euclidean space consists of the ordered set of n distinct vectors

$$\{\mathbf{e}_i : 1 \leq i \leq n\}$$

where $\mathbf{e}_i$ is the ith basis vector, that is, it has a one in the ith coordinate (position) and zeros everywhere else.[1] The $\mathbf{e}_i$ have column vector format

[1] This is also called a "one-hot" encoding in machine learning parlance, where the i's might be the classes in a classification problem.

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \ldots, \mathbf{e}_{n-1} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

Using these definitions we can define the parametric form of some curve C in $\mathbb{R}^3$, $\mathbf{r}(t)$, as follows:

$$\mathbf{r}(t) = g(t)\hat{\mathbf{i}} + h(t)\hat{\mathbf{j}} + k(t)\hat{\mathbf{k}}$$

where $t \in [a, b]$ and $g(t)$, $h(t)$, and $k(t)$ are scalar functions[2].

Another common notation for vectors is $\vec{r}(t)$, where $\vec{r}(t)$ might be defined as follows:

$$\vec{r}(t) = g(t)\hat{i} + h(t)\hat{j} + k(t)\hat{k}$$

In this document I'll use these notations interchangeably.

3 Definitions and a Few Theorems

Definition 3.1. Dot Product: The dot product (aka scalar product or inner product) of two n-dimensional vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$ is defined to be the scalar value

$$\mathbf{a} \cdot \mathbf{b} = \langle \mathbf{a}, \mathbf{b} \rangle = \sum_{j=1}^{n} a_j b_j$$

If $\mathbf{a}$ and $\mathbf{b}$ are column vectors then the dot product can also be written as a matrix product: $\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^\intercal \mathbf{b}$ where $\mathbf{v}^\intercal$ is the transpose of some vector $\mathbf{v}$. Similarly, if $\mathbf{a}$ and $\mathbf{b}$ are row vectors then the dot product can be written as $\mathbf{a} \cdot \mathbf{b} = \mathbf{a}\mathbf{b}^\intercal$.

According to the Pythagorean theorem, the length of a vector $\mathbf{a} = (a_1, a_2, a_3) \in \mathbb{R}^3$, denoted $\|\mathbf{a}\|$, equals $\sqrt{a_1^2 + a_2^2 + a_3^2}$. The next definition is a generalization of the notion of length to vectors in $\mathbb{R}^n$.

Definition 3.2. Euclidean Norm: The Euclidean norm of a vector

$$\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathbb{R}^n$$

is defined to be

$$\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \sqrt{\sum_{j=1}^{n} a_j^2} \tag{1}$$

[2] A scalar function is a function f such that $f : \mathbb{R}^n \to \mathbb{R}$.

Theorem 3.1. Cauchy–Schwarz Inequality:

$$\left|\langle \mathbf{a}, \mathbf{b} \rangle\right| \leq \|\mathbf{a}\|\|\mathbf{b}\|$$

Proof: The inequality is trivial if either $\mathbf{a}$ or $\mathbf{b}$ is zero, so assume that neither is. So without loss of generality let $\mathbf{x} = \dfrac{\mathbf{a}}{\|\mathbf{a}\|}$ and $\mathbf{y} = \dfrac{\mathbf{b}}{\|\mathbf{b}\|}$ and so $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$ ($\mathbf{x} = \hat{\mathbf{a}}$ and $\mathbf{y} = \hat{\mathbf{b}}$).

Then one way to prove the Cauchy–Schwarz inequality is to use the reverse triangle inequality:

$$\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \leq \|\mathbf{x} - \mathbf{y}\| \tag{2}$$

We can get the reverse triangle inequality from the triangle inequality [5, 8] by observing that

$$\begin{aligned}
\|\mathbf{x}\| &= \|\mathbf{x} + (-\mathbf{y} + \mathbf{y})\| && \text{\# add 0 to } \mathbf{x}\text{: } \mathbf{x} = \mathbf{x} + 0 = \mathbf{x} + (-\mathbf{y} + \mathbf{y}) \\
&= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| && \text{\# addition is associative: } \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3) = (\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 \\
&\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| && \text{\# by the triangle inequality: } \|\mathbf{v}_1 + \mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\| \\
&\Rightarrow \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| && \text{\# combine expressions for } \|\mathbf{x}\| \\
&\Rightarrow \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| && \text{\# subtract } \|\mathbf{y}\| \text{ from both sides}
\end{aligned}$$

and

$$\begin{aligned}
\|\mathbf{y}\| &= \|\mathbf{y} + (-\mathbf{x} + \mathbf{x})\| && \text{\# add 0 to } \mathbf{y}\text{: } \mathbf{y} = \mathbf{y} + 0 = \mathbf{y} + (-\mathbf{x} + \mathbf{x}) \\
&= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| && \text{\# addition is associative} \\
&\leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| && \text{\# by the triangle inequality} \\
&\Rightarrow \|\mathbf{y}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| && \text{\# combine expressions for } \|\mathbf{y}\| \\
&\Rightarrow \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| && \text{\# subtract } \|\mathbf{x}\| \text{ from both sides} \\
&\Rightarrow -\Big[\|\mathbf{y}\| - \|\mathbf{x}\|\Big] \geq -\|\mathbf{y} - \mathbf{x}\| && \text{\# multiply both sides by -1: } -1 * (\mathbf{v}_1 \leq \mathbf{v}_2) \Rightarrow -\mathbf{v}_1 \geq -\mathbf{v}_2 \\
&\Rightarrow \|\mathbf{x}\| - \|\mathbf{y}\| \geq -\|\mathbf{x} - \mathbf{y}\| && \text{\# } \|\mathbf{y} - \mathbf{x}\| = \|-(\mathbf{y} - \mathbf{x})\| = \|\mathbf{x} - \mathbf{y}\| \text{ by Equation (1)} \\
&\Rightarrow -\|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x}\| - \|\mathbf{y}\| && \text{\# rearrange}
\end{aligned}$$

Combining these expressions for $\|\mathbf{x}\| - \|\mathbf{y}\|$ gives us the reverse triangle inequality (Equation (2)):

$$-\|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| \Rightarrow \big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \leq \|\mathbf{x} - \mathbf{y}\|$$

To see how the reverse triangle inequality and our definitions of $\mathbf{x}$ and $\mathbf{y}$ imply the Cauchy–Schwarz inequality consider that

$$
\begin{aligned}
\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| &\leq \|\mathbf{x}-\mathbf{y}\| && \text{\# reverse triangle inequality (Equation (2))} \\
&\Rightarrow 0 \leq \|\mathbf{x}-\mathbf{y}\| && \text{\# } \|\mathbf{x}\| = \|\mathbf{y}\| = 1 \text{ by definition above, } 1-1=0,\ |0|=0 \\
&\Rightarrow 0 \leq \|\mathbf{x}-\mathbf{y}\|^2 && \text{\# } \|\mathbf{x}-\mathbf{y}\| \geq 0 \Rightarrow \|\mathbf{x}-\mathbf{y}\|^2 \geq 0 \\
&\Rightarrow 0 \leq \sum_{j=1}^{n}(x_j - y_j)^2 && \text{\# } \|\mathbf{x}-\mathbf{y}\|^2 = \sum_{j=1}^{n}(x_j-y_j)^2 \text{ by Equation (1)} \\
&\Rightarrow 0 \leq \sum_{j=1}^{n}\left(x_j^2 - 2x_jy_j + y_j^2\right) && \text{\# expand previous line} \\
&\Rightarrow 0 \leq \sum_{j=1}^{n} x_j^2 - 2\sum_{j=1}^{n} x_jy_j + \sum_{j=1}^{n} y_j^2 && \text{\# sum is a linear operator} \\
&\Rightarrow 0 \leq \|\mathbf{x}\|^2 - 2\langle\mathbf{x},\mathbf{y}\rangle + \|\mathbf{y}\|^2 && \text{\# } \|\mathbf{x}\|^2 = \sum_{j=1}^{n} x_j^2 \text{ and } \langle\mathbf{x},\mathbf{y}\rangle = \sum_{j=1}^{n} x_jy_j \text{ by Equation (1)} \\
&\Rightarrow 0 \leq 1 - 2\langle\mathbf{x},\mathbf{y}\rangle + 1 && \text{\# } \|\mathbf{x}\| = \|\mathbf{y}\| = 1 \Rightarrow \|\mathbf{x}\|^2 = \|\mathbf{y}\|^2 = 1 \\
&\Rightarrow 0 \leq 2 - 2\langle\mathbf{x},\mathbf{y}\rangle && \text{\# } 1+1=2 \\
&\Rightarrow -2 \leq -2\langle\mathbf{x},\mathbf{y}\rangle && \text{\# subtract 2 from both sides} \\
&\Rightarrow 1 \geq \langle\mathbf{x},\mathbf{y}\rangle && \text{\# divide by -2, reversing inequality} \\
&\Rightarrow \langle\mathbf{x},\mathbf{y}\rangle \leq 1 && \text{\# rearrange} \\
&\Rightarrow \left\langle \frac{\mathbf{a}}{\|\mathbf{a}\|}, \frac{\mathbf{b}}{\|\mathbf{b}\|} \right\rangle \leq 1 && \text{\# } \mathbf{x} = \frac{\mathbf{a}}{\|\mathbf{a}\|},\ \mathbf{y} = \frac{\mathbf{b}}{\|\mathbf{b}\|} \text{ by definition above} \\
&\Rightarrow \frac{\langle\mathbf{a},\mathbf{b}\rangle}{\|\mathbf{a}\|\|\mathbf{b}\|} \leq 1 && \text{\# scalar multiplication rule: } \langle c_1\mathbf{v}_1, c_2\mathbf{v}_2\rangle = c_1c_2\langle\mathbf{v}_1,\mathbf{v}_2\rangle \\
&\Rightarrow \langle\mathbf{a},\mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\| && \text{\# multiply both sides by } \|\mathbf{a}\|\|\mathbf{b}\|
\end{aligned}
$$

If we replace $\mathbf{a}$ by $-\mathbf{a}$ we see by Equation (1) that $\|-\mathbf{a}\| = \|\mathbf{a}\|$. Further, by the scalar multiplication rule for dot products with $c_1 = -1$ and $c_2 = 1$ we have $\langle -\mathbf{a}, \mathbf{b}\rangle = -\langle\mathbf{a},\mathbf{b}\rangle$. Taken together these results imply that[3]

$$-\langle\mathbf{a},\mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$$

Next note that the absolute value of x, $|x|$, is defined to be

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0. \end{cases} \tag{3}$$

So one way to see the Cauchy–Schwarz inequality is by setting x equal to $\langle\mathbf{a},\mathbf{b}\rangle$ in Equation (3). Then when $\langle\mathbf{a},\mathbf{b}\rangle \geq 0$ we have $\big|\langle\mathbf{a},\mathbf{b}\rangle\big| = \langle\mathbf{a},\mathbf{b}\rangle$ and we showed above that $\langle\mathbf{a},\mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$. Similarily, when $\langle\mathbf{a},\mathbf{b}\rangle < 0$ we have $\big|\langle\mathbf{a},\mathbf{b}\rangle\big| = -\langle\mathbf{a},\mathbf{b}\rangle$ and we showed that $-\langle\mathbf{a},\mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$. Together these give us $\big|\langle\mathbf{a},\mathbf{b}\rangle\big| \leq \|\mathbf{a}\|\|\mathbf{b}\|$, namely, the Cauchy–Schwarz inequality. □

[3] There is nothing special about using $\mathbf{a}$ here. We could use $\mathbf{b}$ and get the same result.

3.1 Derivatives

We will also denote the derivative of a vector $\mathbf{r}$ with respect to t by $\dot{\mathbf{r}}$, $\dfrac{d\mathbf{r}}{dt}$ or $\mathbf{r}'(t)$. Note also that by definition

$$\frac{d\mathbf{r}}{dt} = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

3.2 Vector Fields

In vector calculus and physics, a vector field is an assignment of a vector to each point in a subset of space [3]. For example, a vector field in the plane can be visualized as a collection of arrows with a given magnitude and direction, each attached to a point in the plane. Vector fields are often used to model the speed and direction of a moving fluid throughout space, or the strength and direction of some force, such as the magnetic or gravitational force, as it changes from one point to another point.

For example, the wind velocity vector field for the 2011 Joplin, MO tornado [6] is shown in Figure 1. Here the color represents the wind speed $\|\mathbf{v}\|$, where $\mathbf{v}$ is the wind velocity vector.

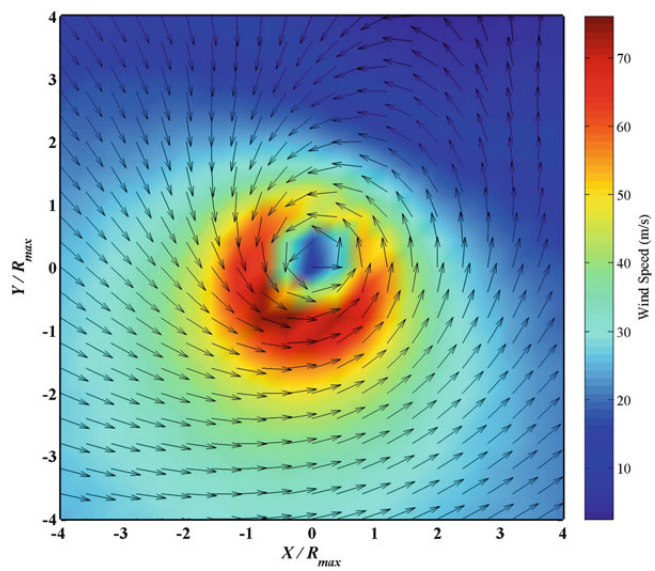

Figure 1: 2011 Joplin, MO Tornado Wind Velocity Vector Field

The general form of a vector field (here in three dimensions) is

$$\vec{F}(x, y, z) = P(x, y, z)\,\hat{\imath} + Q(x, y, z)\,\hat{\jmath} + R(x, y, z)\,\hat{k}$$

where P, Q, and R are scalar functions.

4 Arc Length

Consider a segment of a parametric curve $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}} + k(t)\,\hat{\mathbf{k}}$ between two points $P = \mathbf{r}(t)$ and $Q = \mathbf{r}(t + \Delta t)$, as shown in Figure 2.

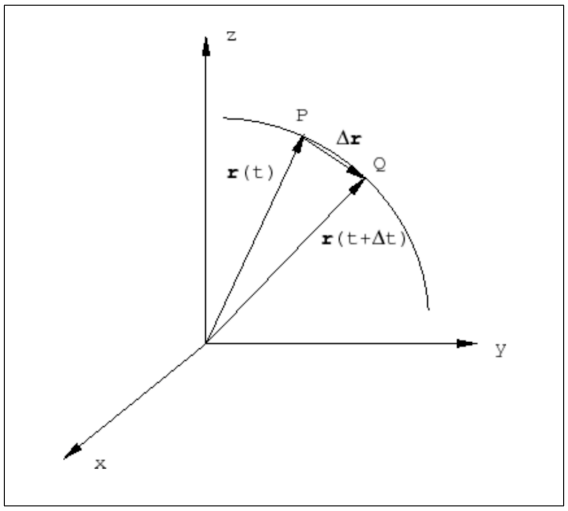

Figure 2: $\Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$

What is the length of the segment of the curve between P and Q? This length is called the arc length and is denoted by Δs. Δs can be approximated by the chord length $\|\Delta\mathbf{r}\|$. Then we can compute the infinitesimal arc length ds as follows:

$$
\begin{aligned}
\Delta s &\approx \|\Delta\mathbf{r}\| && \text{\# approximate } \Delta s \text{ by } \|\Delta\mathbf{r}\| \\
&= \|\mathbf{r}(t+\Delta t) - \mathbf{r}(t)\| && \text{\# } \Delta\mathbf{r} = \mathbf{r}(t+\Delta t) - \mathbf{r}(t) \\
&= \|\mathbf{r}(t+\Delta t) - \mathbf{r}(t)\| \cdot \frac{\Delta t}{\Delta t} && \text{\# multiply by } 1 = \frac{\Delta t}{\Delta t} \\
&= \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# assume } \Delta t > 0 \\
&\Rightarrow \Delta s \approx \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# combine expressions for } \Delta s \\
&\Rightarrow \lim_{\Delta t \to 0} \Delta s = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# take the limit of both sides} \\
&\Rightarrow ds = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# } \Delta t \to 0 \Rightarrow \Delta s \to 0 \Rightarrow \lim_{\Delta t \to 0} \Delta s = ds \\
&\Rightarrow ds = \lim_{\Delta t \to 0} \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# product rule for limits [1]} \\
&\Rightarrow ds = \left\| \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# for a normed vector space } X \ \lim \|x_n\| = \|\lim x_n\| \\
&\Rightarrow ds = \left\| \frac{d\mathbf{r}}{dt} \right\| \cdot dt && \text{\# } \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt}, \ \lim_{\Delta t \to 0} \Delta t = dt
\end{aligned}
$$

Interestingly we can also see this using the Pythagorean theorem. Specifically, if we move an infinitesimal distance in the x and y directions we get a triangle in the x-y plane who's sides have length dx and dy and who's hypotenuse is ds. This scenario is shown for some curve $f(x)$ in Figure 3.

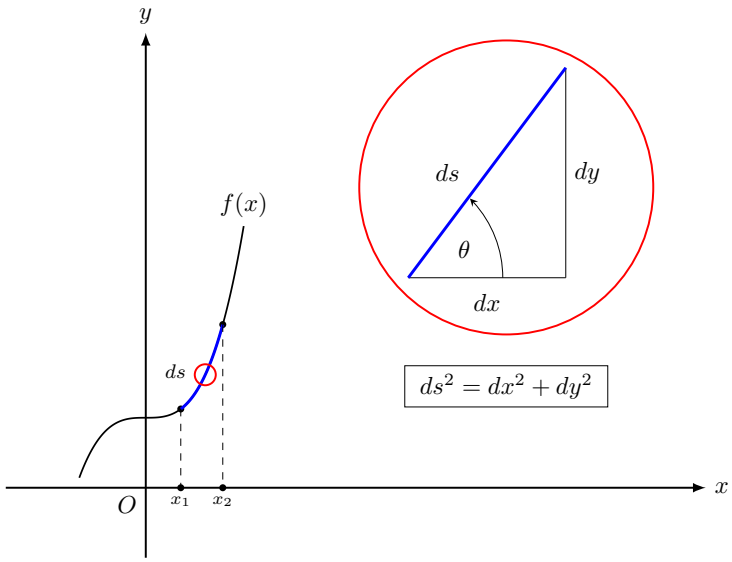

Figure 3: $f(x)$, ds and the Pythagorean Theorem

In the two dimensional case where the parameterization is $x = g(t)$ and $y = h(t)$ we have $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}}$. Further, since derivative is a linear operator [7] we know that

$$\frac{d\mathbf{r}}{dt} = \frac{dg}{dt}\,\hat{\mathbf{i}} + \frac{dh}{dt}\,\hat{\mathbf{j}} \tag{4}$$

So in the source domain the Pythagorean theorem tells us that $ds = \sqrt{dx^2 + dy^2}$. Using this expression for ds we can see that

$$\begin{aligned}
ds &= \sqrt{dx^2 + dy^2} && \text{\# Figure 3} \\
&= \sqrt{dg^2 + dh^2} && \text{\# switch to parametric form: } x = g(t) \text{ and } y = h(t) \\
&= \sqrt{dg^2 + dh^2} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt},\ dt > 0 \\
&= \sqrt{dg^2 + dh^2} \cdot \left[\sqrt{\frac{1}{dt^2}} \cdot dt\right] && \text{\# } \frac{x}{x} = \sqrt{\left(\frac{1}{x}\right)^2} \cdot x = \sqrt{\frac{1}{x^2}} \cdot x,\ x > 0 \\
&= \sqrt{\left(dg^2 + dh^2\right) \cdot \left(\frac{1}{dt^2}\right)} \cdot dt && \text{\# simplify} \\
&= \sqrt{\left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2} \cdot dt && \text{\# simplify} \\
&= \sqrt{g'(t)^2 + h'(t)^2} \cdot dt && \text{\# switch notation from Leibniz} \rightarrow \text{Lagrange: } \frac{df}{dt} = f'(t) \\
&= \left\|\frac{d\mathbf{r}}{dt}\right\| \cdot dt && \text{\# Equation (4) and the definition of } \|\cdot\|
\end{aligned}$$

In any event, now we see that

$$\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\| \tag{5}$$

and so

$$\frac{ds}{dt} = \|\dot{\mathbf{r}}\| \tag{6}$$

Since $\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\| = \|\dot{\mathbf{r}}\|$ we also know that the unit tangent vector $\mathbf{T} = \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} = \frac{d\mathbf{r}}{ds}$ [4]. One way to see this is to notice that

$$\begin{aligned}
\mathbf{T} &= \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} && \text{\# definition of the unit tangent vector } \mathbf{T} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\|\dot{\mathbf{r}}\|} && \text{\# definition of } \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} && \text{\# Equation (6): } \|\dot{\mathbf{r}}\| = \frac{ds}{dt} \\
&= \frac{d\mathbf{r}}{ds} && \text{\# simplify}
\end{aligned}$$

5 Conclusions

Acknowledgements

LaTeX Source

References

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